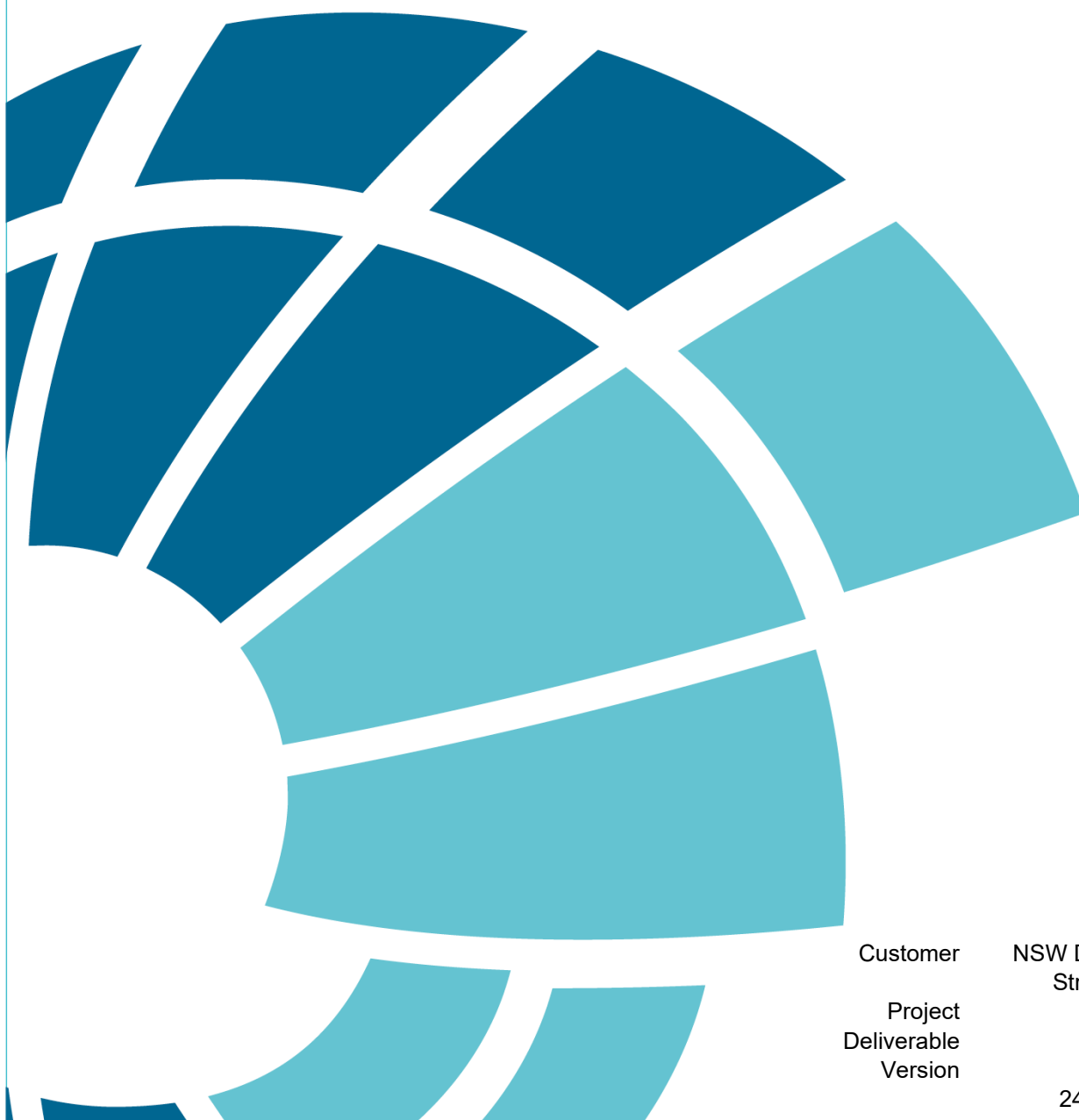


Old Castlereagh Road Water Quality Impact Assessment

Mass Balance Study for Due Dilligence



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Executive Summary

NSW Department of Planning, Housing and Infrastructure (DPHI) commissioned BMT to undertake a due diligence assessment of the potential water quality impacts associated with the proposed Old Castlereagh Road development adjacent to the Penrith Lakes Scheme. The assessment considered whether stormwater runoff generated by the development would result in a material change in water quality within the Final Basin and downstream Sydney International Regatta Centre (Regatta) lake system. The assessment was based on available water quality monitoring records, water level data, lake storage estimates, and stormwater runoff and pollutant load information from existing flood and MUSIC modelling.

The assessment was undertaken using a screening-level mass balance approach to compare existing and post-development conditions for Total Nitrogen (TN), Total Phosphorus (TP), Total Suspended Solids (TSS) and Chlorophyll-a (Chl-a). Existing conditions indicate that the Final Basin already experiences elevated nutrient concentrations, with TP and Chl-a routinely exceeding adopted Water Quality Objectives, while seasonal TN exceedances also occur. Water quality within Regatta is generally better than the Final Basin, although periodic phosphorus and Chl-a exceedances are observed under existing conditions.

Development-derived stormwater loads were found to be small relative to the existing constituent mass inventories of the receiving waterbodies. Predicted changes in TN, TP and TSS concentrations within the Final Basin were negligible and did not materially alter compliance outcomes or seasonal water quality patterns. Event-based assessments for the 1% and 0.5% AEP rainfall events indicated that additional pollutant loads represent less than 1% of the existing constituent mass inventory within the Final Basin.

Assessment of potential overflow from the Final Basin to Regatta demonstrated that only small constituent loads are transferred downstream during high-water-level periods. The transferred loads were negligible relative to the large storage volume and existing constituent masses within Regatta, resulting in no material change to predicted water quality conditions or compliance outcomes.

Overall, the assessment concludes that the proposed development is unlikely to result in a material deterioration of water quality within either the Final Basin or Regatta. Existing exceedances of nutrient-related WQOs are expected to remain substantially unchanged, while TSS is predicted to remain compliant. Given the conservative assumptions adopted, including the use of untreated MUSIC source loads, the assessment is considered to provide a precautionary estimate of potential impacts.

This study was undertaken in accordance with the National Health and Medical Research Council (NHMRC) Guidelines for Managing Risks in Recreational Water (2008). Under the NHMRC guidance, the applicable water quality objectives were considered to assess the relative impact of predicted water quality changes associated with the proposed development.

Glossary of Acronyms

Acronym	Definition
AEP	Annual Exceedance Probability
AHD	Australian Height Datum
BMT	BMT Commercial Australia Pty Ltd
Chl-a	Chlorophyll-a
DPHI	Department of Planning, Housing and Infrastructure (NSW)
ML	Megalitre
MUSIC	Model for Urban Stormwater Improvement Conceptualisation
NHMRC	National Health and Medical Research Council
SIRC	Sydney International Regatta Centre
TN	Total Nitrogen
TP	Total Phosphorus
TSS	Total Suspended Solids
WMP	Water Management Plan
WQO	Water Quality Objective
WQ	Water Quality

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1 Introduction

The proposed Old Castlereagh Road development is located adjacent to the Penrith Lakes Scheme and will generate additional stormwater runoff from proposed tourism, recreation and commercial land uses. NSW Department of Planning, Housing and Infrastructure (DPHI) has commissioned a due diligence assessment to determine whether development-derived stormwater loads are likely to result in a material change in water quality within the receiving lake system.

Stormwater runoff from the development is expected to discharge to the Final Basin, which forms part of the eastern lake network within the Penrith Lakes Scheme. During periods of elevated water level, water stored within the Final Basin may be transferred to the downstream Sydney International Regatta Centre (SIRC) lake system via existing hydraulic connections. The SIRC lake system comprises the hydraulically connected Rowing Lake and Warm Up Lake, which are assessed collectively in this report and referred to as the Regatta. Consequently, the assessment considers both the direct effects of development-derived stormwater loads on the Final Basin and the potential downstream implications for Regatta.

This technical note presents a screening-level water quality assessment based on available monitoring data, water level records, waterbody storage estimates and stormwater quality modelling outputs. The assessment applies a simplified mass-balance approach to compare existing and post-development conditions and determine whether the proposed development is likely to materially affect water quality within the receiving waterbodies.

1.1 Objectives

The overall objective of this assessment is to determine whether stormwater runoff generated by the proposed development is likely to result in a material change in water quality within the Final Basin and downstream Regatta relative to existing conditions and adopted water quality objectives.

To achieve this objective, the assessment:

- Characterises existing water quality conditions within the Final Basin and Regatta.
- Quantifies development-derived stormwater loads entering the Final Basin.
- Assesses the potential effect of additional loads on constituent mass and concentrations within the Final Basin.
- Assesses the potential transfer of water and constituent mass from the Final Basin to Regatta under assumed overflow conditions.
- Compares existing and post-development concentrations against adopted water quality objectives (WQOs) to determine the significance of any predicted change

1.2 Assessment Framework

The assessment is based on a simplified conceptual model in which development-derived stormwater runoff enters the Final Basin and becomes mixed within the receiving waterbody. Under elevated water level conditions, a proportion of stored water may be transferred from the Final Basin to Regatta

through operation of the existing lake system. Any constituent mass associated with transferred water is assumed to be conveyed downstream and incorporated into the Regatta mass balance.

Accordingly, two receiving waterbodies are considered:

- Final Basin – the primary receiving waterbody for development-derived stormwater runoff.
- Regatta – the downstream receptor that may receive transferred water and constituent mass from the Final Basin during overflow events.

Water quality impacts are assessed by comparing constituent mass inventories and concentrations under existing and post-development conditions within both waterbodies. The assessment focuses on Total Nitrogen (TN), Total Phosphorus (TP), Total Suspended Solids (TSS) and Chlorophyll-a (Chl-a), consistent with the available monitoring data and adopted WQOs

1.3 Scope of Assessment

This assessment includes:

- Review of available water quality and water level monitoring data.
- Characterisation of existing water quality conditions within the Final Basin and Regatta.
- Estimation of existing constituent mass inventories using monitored concentrations and corresponding waterbody storage volumes.
- Estimation of development-derived stormwater runoff volumes and pollutant loads using available flood and MUSIC modelling outputs.
- Assessment of potential water and constituent mass transfers from the Final Basin to Regatta.
- Estimation of post-development constituent concentrations using a simplified mass-balance approach.
- Comparison of existing and post-development conditions against adopted water quality objectives.

The assessment should therefore be regarded as a screening-level due diligence assessment intended to evaluate the relative magnitude of development-derived stormwater loads compared with existing water quality conditions within the receiving lake system.

The assessment does not include detailed hydrological, hydrodynamic or ecological process modelling.

2 Data Sources

This assessment is based on three principal sources of information:

- Water quality and water level monitoring data used to characterise existing conditions within the Final Basin and Regatta.
- Water level-storage relationships used to estimate waterbody storage volumes and constituent mass inventories.
- Flood and stormwater quality modelling outputs used to estimate development-derived runoff volumes and pollutant loads.

Together, these datasets provide the basis for the screening-level mass-balance assessment described in Section 3.

2.1 Monitoring Data

Water quality and water level monitoring data were provided by NSW DPHI for the Final Basin and Regatta. These datasets were used to characterise existing water quality conditions, estimate waterbody storage volumes and calculate constituent mass inventories for each waterbody. Available monitoring parameters included total nitrogen (TN), total phosphorus (TP), total suspended solids (TSS), chlorophyll-a (Chl-a) and water level. A summary of the available monitoring record is provided in Table 2.1.

Table 2.1 Monitoring Data Summary

Dataset	Final Basin	Regatta	Remarks
TN, TP, TSS	Nov 2022 - Jun 2026	Nov 2022 - Jun 2026	Used for existing condition assessment
Chlorophyll-a	Nov 2022 - Jan 2025	Nov 2022 - Jan 2025	Monitoring discontinued after Jan 2025
Water level	Jul 2020 - Jun 2026	Jul 2020 - Jun 2025	Minor data gaps present

2.1.2 Final Basin

The Final Basin dataset comprised water quality monitoring data for TN, TP, TSS and Chl-a together with continuous water level records. Water level data were converted to storage volume using the adopted water level-storage relationship, enabling calculation of constituent mass inventories and assessment of temporal changes in water quality conditions.

An overflow threshold of 18.0 m AHD was adopted for this assessment because the Water Management Plan identifies 18.0 m AHD as the upper operational water level for the Final Basin (Water Technology, 2026). Timeseries storage volumes for the Final Basin were estimated by applying the adopted water level-volume relationship to the monitored water level time series, with a storage volume of 75 ML corresponding to the operational water level of 16.0 m AHD (Water Technology, 2026). Representative time series of monitored water quality, water level and derived storage volume are presented in Figure 2.1 to Figure 2.3.

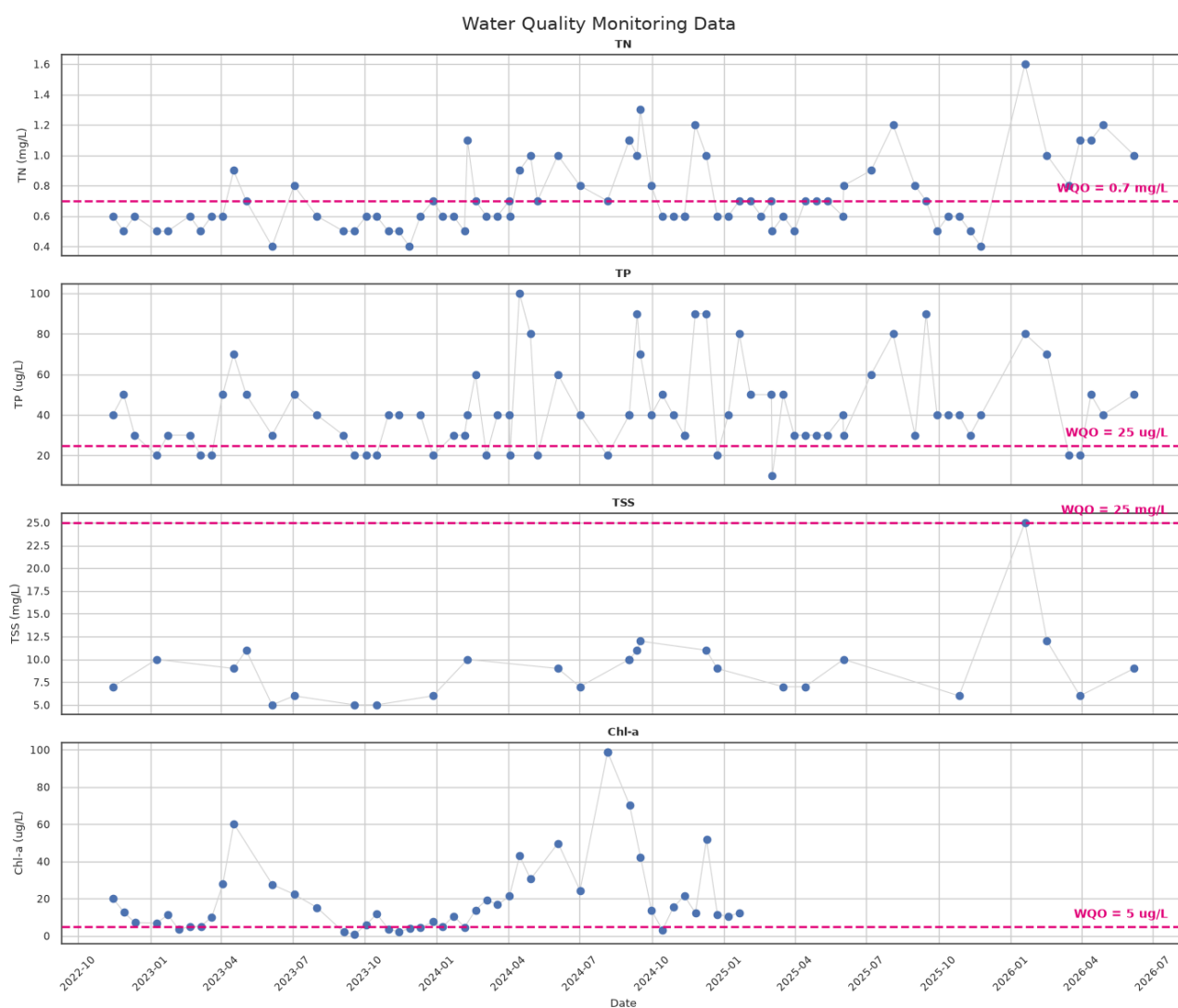


Figure 2.1 Final basin – timeseries water quality (TN, TP, TSS and Chl-a)

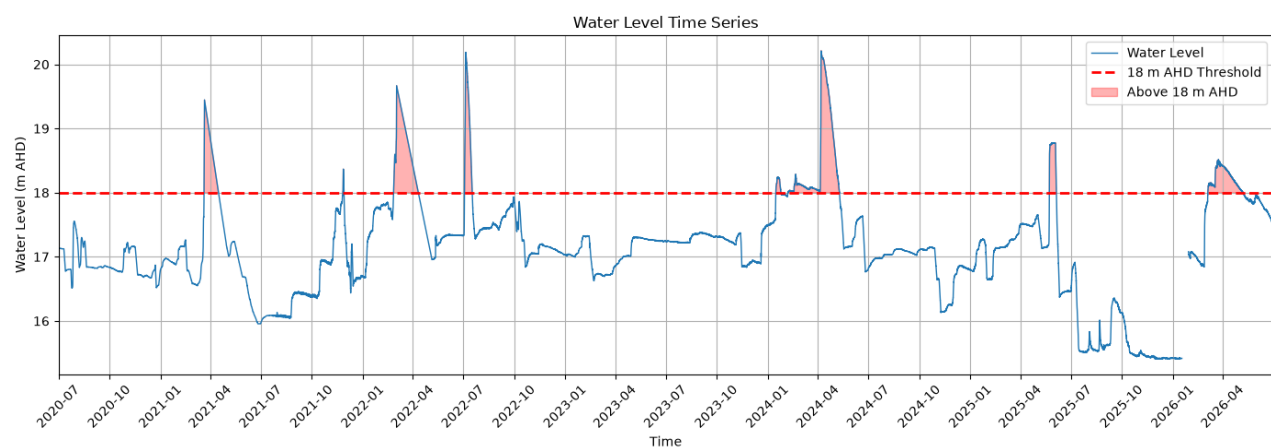


Figure 2.2 Final basin – timeseries water level; 18m represents upper operational level

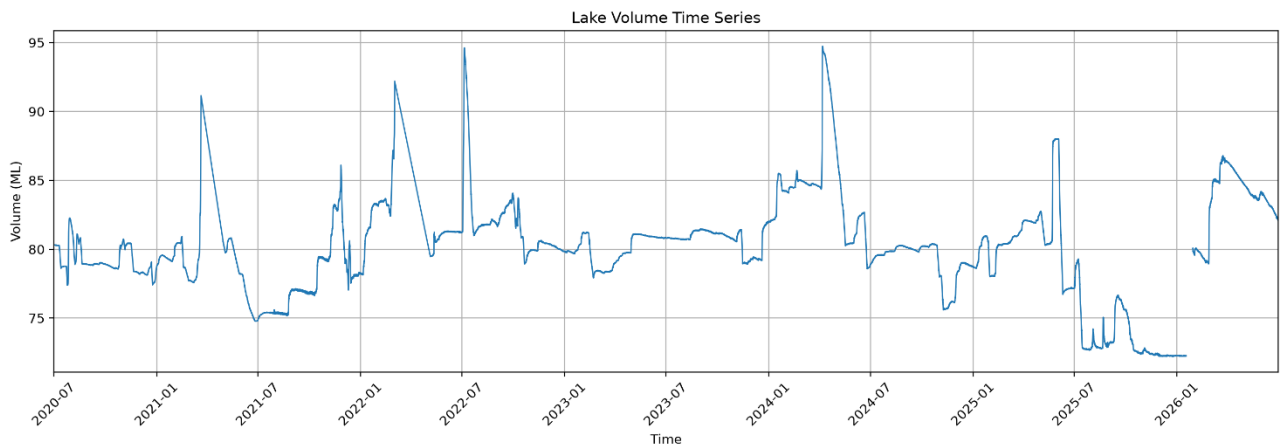


Figure 2.3 Final basin – timeseries derived storage volume

2.1.3 Regatta

The Regatta dataset comprised water quality monitoring data collected throughout the Sydney International Regatta Centre (SIRC) lake system together with continuous water level measurements. Waterbody storage volumes were estimated from the monitored water level time series using the adopted water level-volume relationship, whereby a water level of 15.0 m AHD corresponds to a storage volume of approximately 4,223 ML (Cardno, 2012). Representative water quality, water level and derived storage volume time series are presented in Figure 2.4 to Figure 2.6.

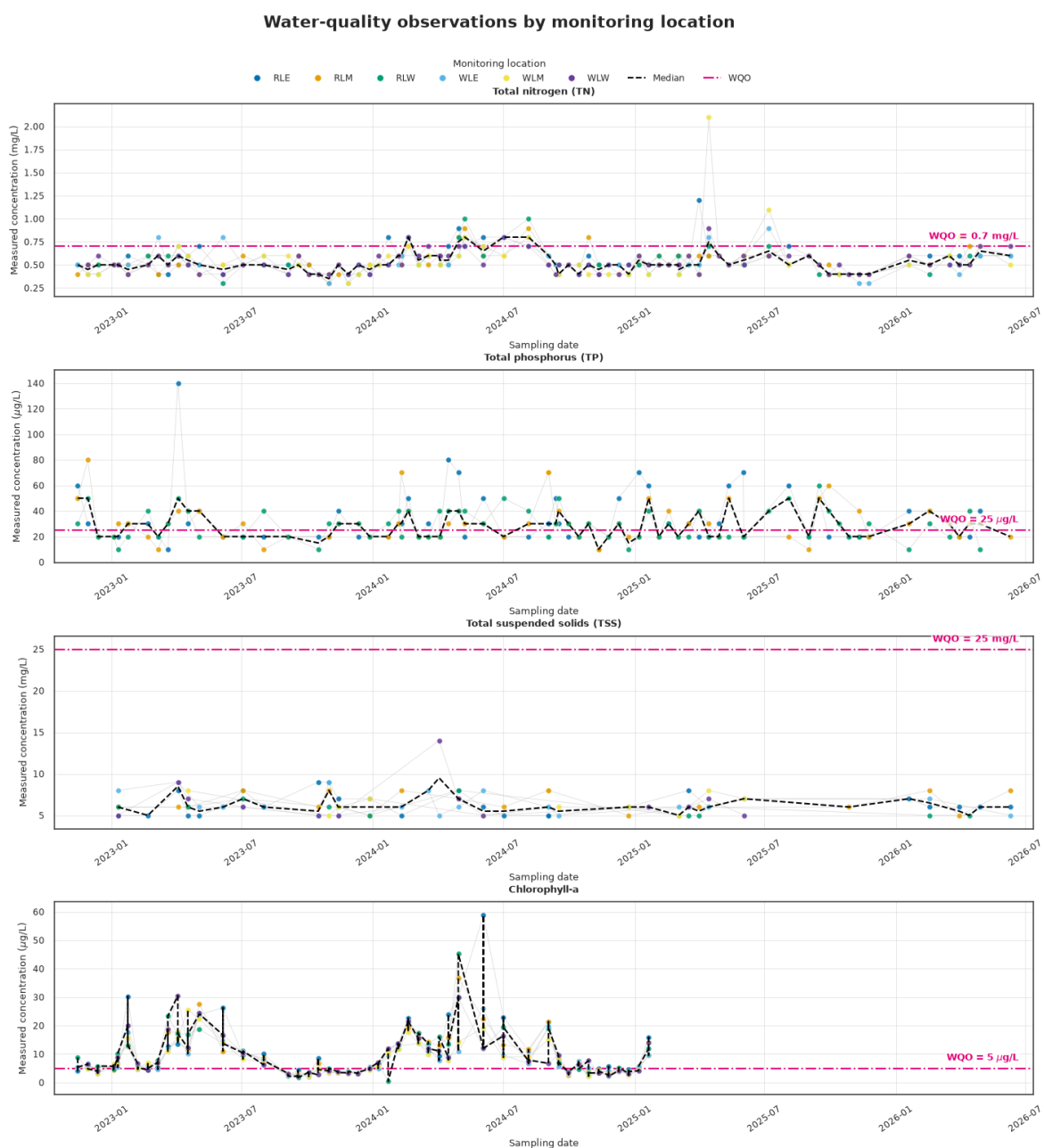


Figure 2.4 Regatta – timeseries water quality (TN, TP, TSS and Chl-a)

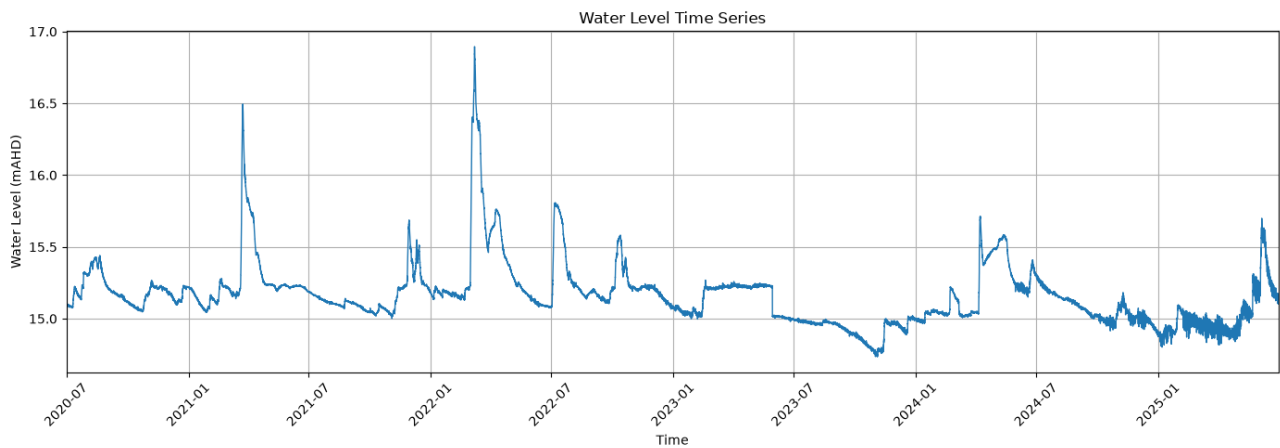


Figure 2.5 Regatta – timeseries water level

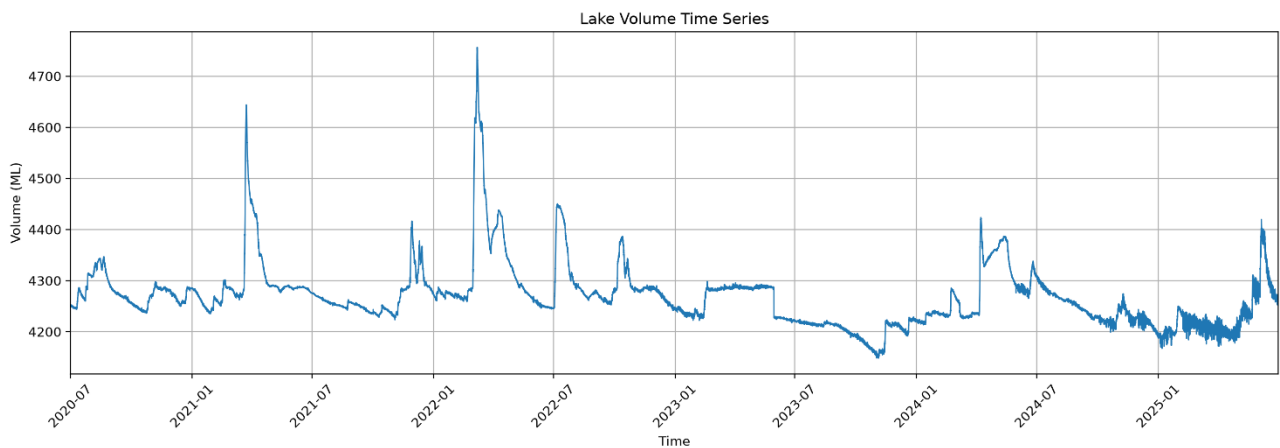


Figure 2.6 Regatta – timeseries derived storage volume

2.1.4 Summary Statistics

Summary statistics derived from the monitored water level and water quality datasets are presented in Table 2.2. The median values provide an indication of typical baseline conditions within each waterbody, while the maximum values represent the upper range of observed conditions during the monitoring period.

Key observations;

- The Final Basin generally exhibits higher nutrient and chlorophyll-a concentrations than Regatta.
- TP concentrations frequently exceed the adopted assessment criterion in both waterbodies.
- Chl-a concentrations are substantially higher in the Final Basin, indicating greater algal productivity.
- Regatta generally exhibits lower nutrient concentrations and improved water quality conditions relative to the Final Basin.

Table 2.2 Monitoring Data - Key Statistics

Variable		Final Basin	Regatta	unit
WL	median	17.11	15.15	m
WL	max	20.21	16.9	m
TN	median	0.6	0.5	mg/L
TN	max	1.6	2.1	mg/L
TP	median	40	30	µg/L
TP	max	100	140	µg/L
TSS	median	9	6	mg/L
TSS	max	25	14	mg/L
Chl-a	median	12.61	6.7	µg /L
Chl-a	max	98.9	58.8	µg /L

2.1.5 Temperature Profile Monitoring Data

Temperature profile monitoring data were reviewed qualitatively to identify seasonal stratification patterns that may influence algal productivity within the Final Basin. The monitoring record indicates a consistent seasonal pattern throughout the assessment period. The water column was generally well mixed during winter and peak summer periods, while thermal stratification was most evident during autumn and spring when distinct vertical temperature gradients developed. Representative examples for the 2020-2021 and 2025-2026 monitoring years are shown in Figure 2.7 and Figure 2.8.

The temperature data are used only as supporting information for interpretation of chlorophyll-a and algal productivity trends and are not directly incorporated into the mass-balance calculations.

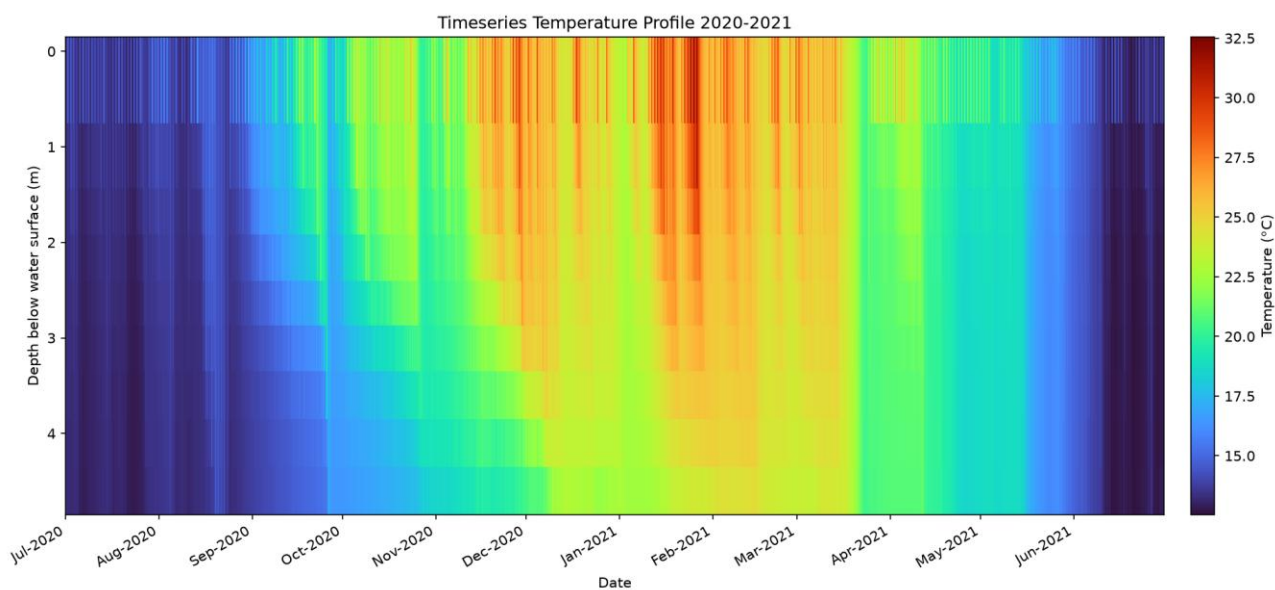


Figure 2.7 Timeseries temperature depth profile from July 2020 to June 2021

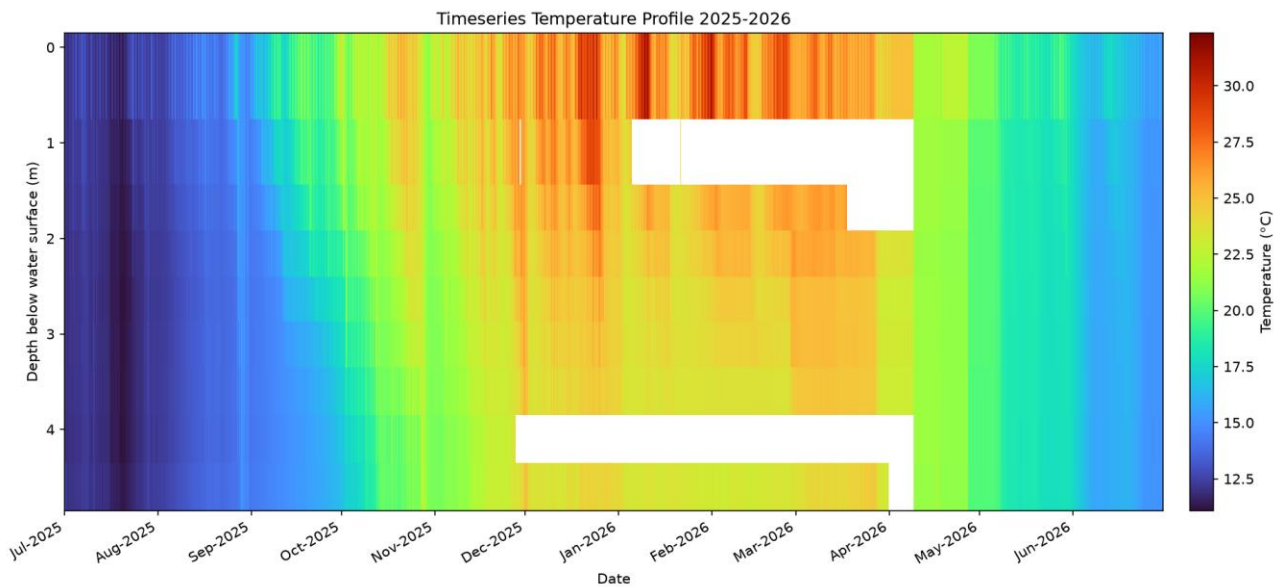


Figure 2.8 Timeseries temperature depth profile from July 2025 to June 2026

2.2 Development Design Inputs

Development-derived runoff volumes and pollutant loads were obtained from existing flood and stormwater quality assessments prepared for the proposed development (Table 2.3 and Table 2.4). These datasets (Smart Structures Australia, 2026; Water Technology, 2026) were used to estimate the volume of additional runoff entering the Final Basin and the associated constituent loads under post-development conditions.

The following information was adopted:

- Post-development runoff volumes for the 1% AEP and 0.5% AEP rainfall events
- Annual stormwater runoff volume derived from MUSIC modelling.
- Annual TN, TP and TSS loads derived from MUSIC modelling.

2.2.1 Conservative Assessment Assumption

A conservative approach was adopted whereby MUSIC source loads were used in the assessment. Source loads represent untreated runoff generated by the proposed development prior to implementation of stormwater treatment measures.

Consequently, the assessment is likely to overestimate actual post-development pollutant loads reaching the receiving waterbodies because potential treatment benefits have not been credited. The resulting impact assessment should therefore be considered conservative.

Table 2.3 Volumes from catchments in the AEP storm event in pre and post-development conditions

AEP	Development scenario	Total Volume (m ³)	Additional storm water volumes(m ³)
1%	Existing	869.4	30.4
	Post-Development	899.8	
0.5%	Existing	1015.2	37
	Post-Development	1052.2	

Table 2.4 MUSIC Model Outputs – Post-development Source Loads

Parameter	Value
Flow (ML/yr)	12.2
Total Suspended Solids (kg/yr)	947
Total Phosphorus (kg/yr)	2.47
Total Nitrogen (kg/yr)	26.8
Gross Pollutants (kg/yr)	323

2.3 Water Quality Assessment Criteria

The assessment of water quality impacts was undertaken using WQOs previously adopted for the Penrith Lakes Scheme. These objectives are derived from the established Penrith Lakes water quality assessment framework, which incorporates the requirements and guidance provided in:

- 1987 Penrith Lakes Scheme Deed of Agreement (DoA)
- Australian and New Zealand Guidelines for Fresh and Marine Water Quality (ANZECC, 2000)
- Guidelines for Managing Risks in Recreational Water (NHMRC, 2008)

The adopted WQOs have been applied in previous assessments of water quality performance within the Penrith Lakes Scheme and provide a consistent basis for evaluating existing and post-development conditions (BMT, 2020; 2022). The assessment criteria adopted in this study are summarised in Table 2.5 and were used to assess whether the proposed development is likely to result in material changes to water quality within the Penrith Lakes Scheme.

Table 2.5 Water Quality Assessment Criteria

Parameter	Assessment criterion	Units
Total Nitrogen (TN)	< 0.7	mg/L
Total Phosphorus (TP)	< 25	µg/L
Total Suspended Solids (TSS)	< 25	mg/L
Chlorophyll-a (Chl-a)	< 5	µg/L
Algal bloom susceptibility (qualitative)	TP < 25 indicates Low to Moderate susceptibility; TP > 25 may indicate increasing susceptibility	µg/L

3 Water Quality Impact Assessment

3.1 Assessment Overview

A simplified mass-balance assessment was undertaken to evaluate the potential effects of development-derived stormwater loads on water quality within the Final Basin and downstream Regatta system. The assessment combines monitoring data, waterbody storage estimates and development-derived runoff and pollutant load data to compare existing and post-development conditions.

The assessment was undertaken in six stages:

1. Characterisation of existing water quality conditions within the Final Basin and Regatta.
2. Quantification of development-derived stormwater loads entering the Final Basin.
3. Assessment of post-development conditions within the Final Basin.
4. Assessment of constituent transfer from the Final Basin to Regatta under assumed overflow conditions.
5. Assessment of post-development conditions within Regatta.
6. Comparison of existing and post-development conditions against adopted WQOs.

The overall assessment workflow is shown in Figure 3.1.

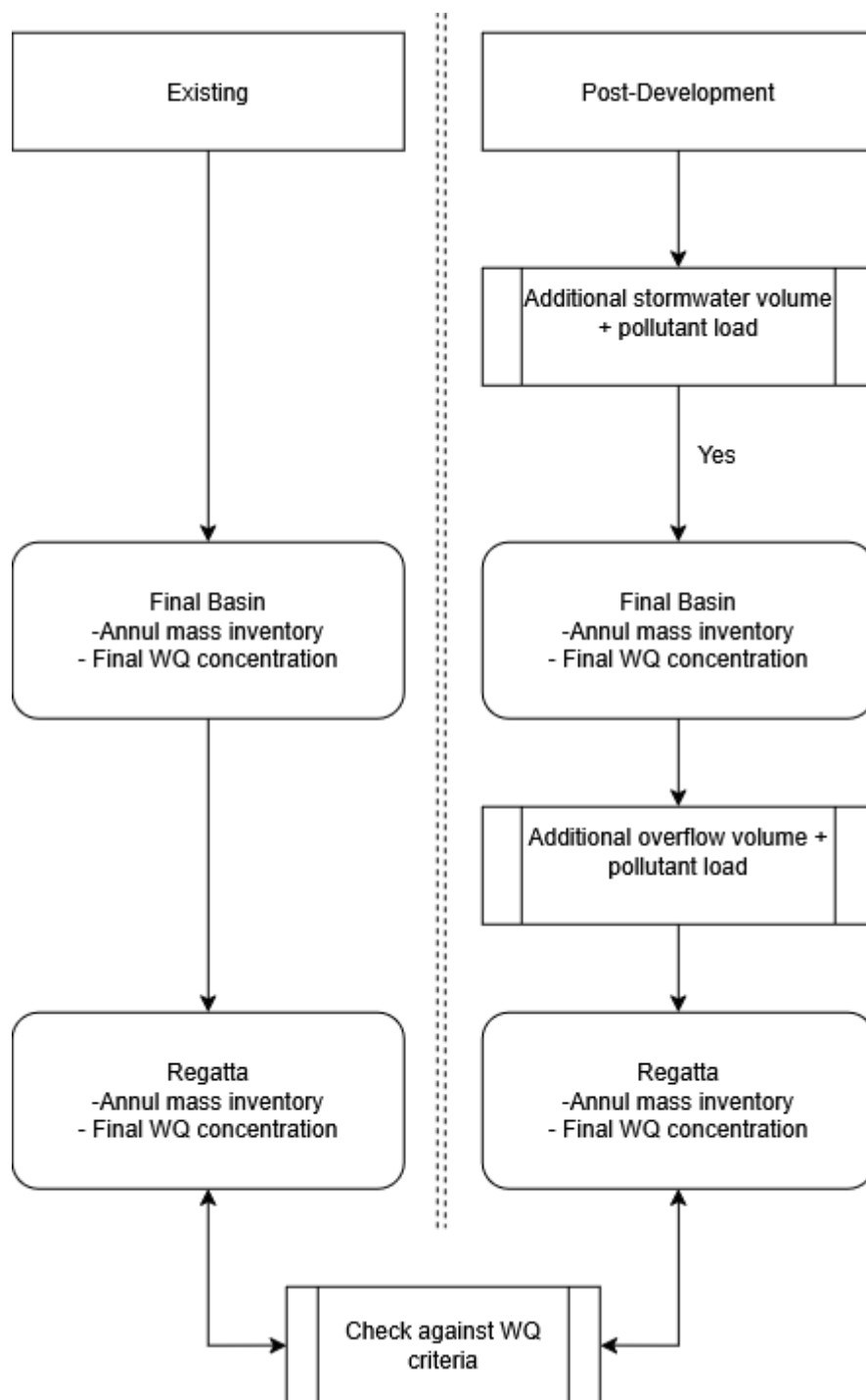


Figure 3.1 Assessment workflow diagram for the due diligence work in the water quality assessment

3.2 Existing Water Quality Assessment

Monthly median concentrations of TN, TP, TSS and Chl-a were calculated for the Final Basin and Regatta and compared against the adopted Penrith Lakes WQOs (Table 3.1 and Table 3.2). This assessment provides the baseline against which potential development impacts are evaluated.

3.2.1 Final Basin

Key findings are:

- TN concentrations exceeded the adopted objective of 0.7 mg/L during much of the cooler period between April and September.
- TP concentrations exceeded the adopted objective of 25 µg/L throughout the year.
- Chl-a concentrations exceeded the adopted objective of 5 µg/L throughout the year.
- TSS concentrations remained below the adopted objective of 25 mg/L throughout the monitoring period.

These results indicate that the Final Basin already experiences elevated nutrient concentrations and persistent exceedances of the adopted TP and Chl-a objectives under existing conditions.

3.2.2 Regatta

Key findings are:

- TN concentrations complied with the adopted objective throughout the year.
- TP concentrations periodically exceeded the adopted objective, particularly during late summer and autumn.
- Chl-a concentrations exceeded the adopted objective during several months but were generally lower than those observed in the Final Basin.
- TSS concentrations remained below the adopted objective throughout the monitoring period.

Overall, Regatta Lake generally exhibits lower nutrient concentrations and improved water quality conditions relative to the Final Basin. The existing condition assessment demonstrates that TP and Chl-a exceedances already occur within the lake system under baseline conditions. These existing conditions provide important context for evaluating the significance of development-derived stormwater loads.

Table 3.1 Monthly median existing concentrations in the Final Basin and comparison against adopted WQOs

Month	TN (mg/L)	Criterion Met?	TP (µg/L)	Criterion Met?	TSS (mg/L)	Criterion Met?	Chl-a (µg/L)	Criterion Met?
Jan	0.63	Y	45	N	17.5	Y	9.1	N
Feb	0.68	Y	45	N	11	Y	6.8	N
Mar	0.58	Y	25	N	6.5	Y	12.8	N
Apr	0.78	N	52.5	N	8	Y	37.28	N
May	0.70	N	30	N	11	Y	37.99	N
Jun	0.85	N	42.5	N	9	Y	38.7	N
Jul	0.80	N	50	N	6.5	Y	23.4	N
Aug	0.70	N	40	N	7.1	Y	57.05	N

Month	TN (mg/L)	Criterion Met?	TP (µg/L)	Criterion Met?	TSS (mg/L)	Criterion Met?	Chl-a (µg/L)	Criterion Met?
Sep	0.70	N	40	N	7.8	Y	21.93	N
Oct	0.60	Y	40	N	5.5	Y	7.75	N
Nov	0.50	Y	37.5	N	7	Y	16.4	N
Dec	0.65	Y	30	N	8	Y	7.1	N

Table 3.2 Monthly median existing concentrations in Regatta lake and comparison against adopted WQOs

Month	TN (mg/L)	Criterion Met?	TP (µg/L)	Criterion Met?	TSS (mg/L)	Criterion Met?	Chl-a (µg/L)	Criterion Met?
Jan	0.5	Y	20	Y	6	Y	7.61	N
Feb	0.5	Y	30	N	5.5	Y	9.42	N
Mar	0.55	Y	25	N	6.75	Y	11.52	N
Apr	0.58	Y	30	N	7.25	Y	14.48	N
May	0.5	Y	40	N	5.5	Y	24.05	N
Jun	0.55	Y	20	Y	6	Y	17.55	N
Jul	0.65	Y	20	Y	6.25	Y	12.62	N
Aug	0.65	Y	25	N	6	Y	8.1	N
Sep	0.49	Y	25	N	5.75	Y	4.85	Y
Oct	0.43	Y	21.25	Y	6.75	Y	4.62	Y
Nov	0.48	Y	30	N	6	Y	3.5	Y
Dec	0.48	Y	22.5	Y	6	Y	4.01	Y

3.3 Post-Development Water Quality Assessment

3.3.1 Development Derived Stormwater Load

Development-derived runoff volumes and pollutant loads were estimated using the available MUSIC model outputs (Table 2.4). A conservative assessment approach was adopted whereby MUSIC source loads were used. These loads represent untreated runoff entering the receiving environment and do not account for pollutant reductions that may be achieved by proposed stormwater treatment measures. Consequently, the assessment is likely to overestimate actual post-development pollutant loads entering the Final Basin.

Annual stormwater runoff volumes and constituent loads were incorporated into the mass-balance assessment to estimate post-development constituent masses (Figure 3.2) and concentrations within the receiving waterbodies.

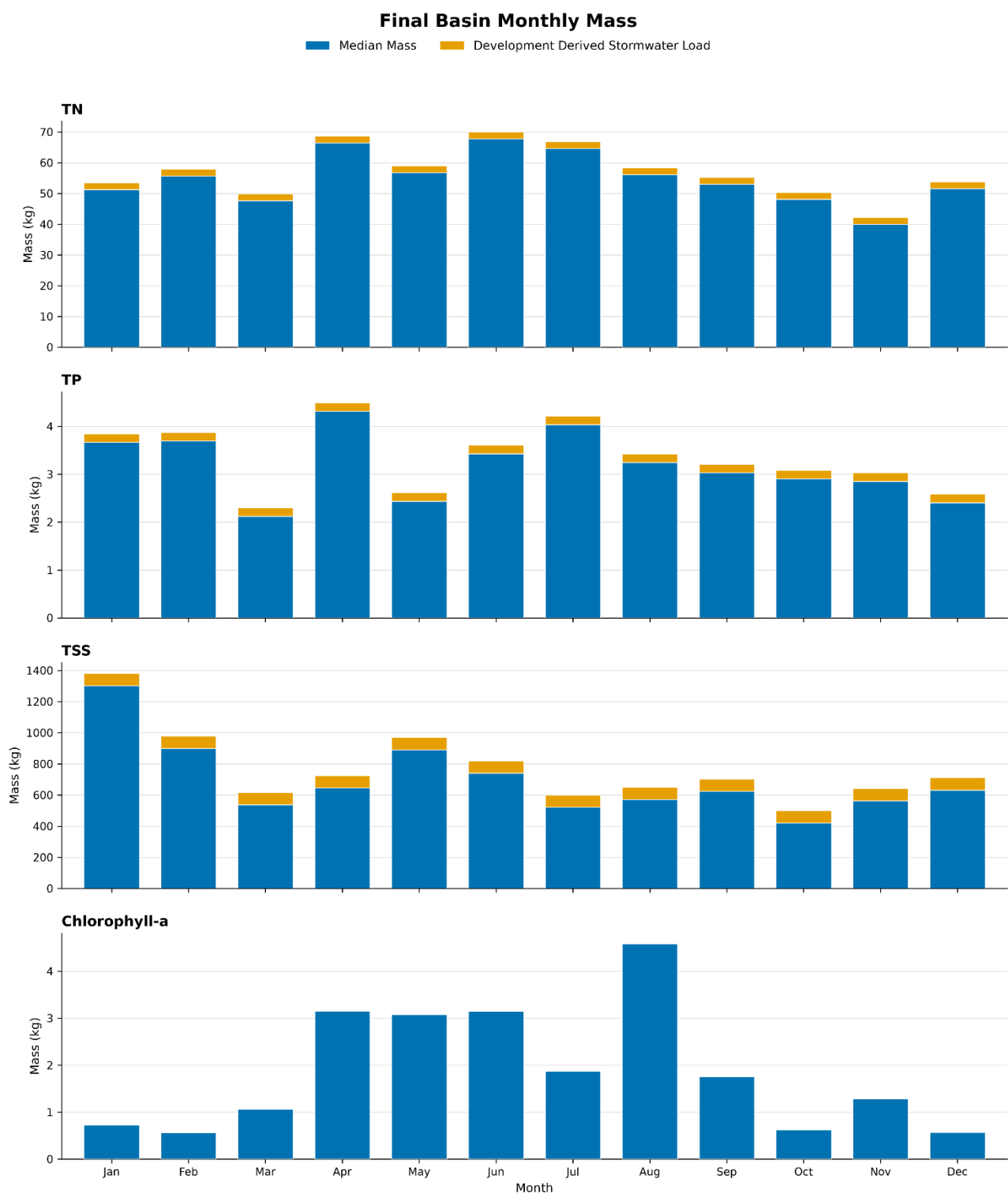


Figure 3.2 Post-development Final Basin constituent mass inventory (Monthly median)

3.3.2 Final Basin Assessment

The assessment indicates that development-derived stormwater loads result in only minor changes to constituent concentrations within the Final Basin and do not materially alter existing compliance outcomes (Table 3.3).

Key findings are:

- TN concentrations remain broadly consistent with existing conditions and continue to exceed the adopted objective during much of the cooler period.
- TP concentrations continue to exceed the adopted objective throughout the year.
- TSS concentrations remain below the adopted objective throughout the year.
- Existing seasonal patterns of water quality remain unchanged.

The assessment therefore indicates that development-derived loads are small relative to the existing constituent mass inventory within the Final Basin.

Table 3.3 Monthly median post-development concentrations in the Final Basin and comparison against adopted WQOs.

Month	TN (mg/L)	Criterion Met?	TP (µg/L)	Criterion Met?	TSS (mg/L)	Criterion Met?
Jan	0.66	Y	47.47	N	17.05	Y
Feb	0.71	N	47.67	N	12.04	Y
Mar	0.60	Y	27.42	N	7.36	Y
Apr	0.81	N	52.98	N	8.54	Y
May	0.72	N	31.82	N	11.82	Y
Jun	0.85	N	43.91	N	9.98	Y
Jul	0.83	N	52.32	N	7.44	Y
Aug	0.72	N	42.17	N	8.01	Y
Sep	0.68	Y	39.57	N	8.67	Y
Oct	0.62	Y	37.96	N	6.15	Y
Nov	0.54	Y	38.50	N	8.15	Y
Dec	0.67	Y	32.23	N	8.88	Y

Post-development Chl-a concentrations were not estimated because development-derived Chl-a loads were unavailable. However, review of the monitoring record indicates that elevated Chl-a concentrations generally occur during periods of enhanced thermal stratification.

Given the relatively small increase in nutrient loads predicted for the development, the proposed development is unlikely to materially alter existing algal bloom risk or the existing seasonal pattern of algal productivity within the Final Basin.

3.3.3 Event-Based Assessment – Final Basin

An event-based assessment was undertaken to evaluate the potential significance of development-derived stormwater loads during the 1% AEP and 0.5% AEP rainfall events (Table 3.4). The assessment indicates that additional constituent loads generated by the development represent less than 1% of the existing constituent mass inventory within the Final Basin for all assessed parameters.

Key findings include:

- TN mass increases of approximately 0.1-0.2%.
- TP mass increases of approximately 0.3%.
- TSS mass increases of approximately 0.3-0.5%.

Even under the larger 0.5% AEP event, development-derived pollutant loads remain very small compared with the existing constituent mass inventory of the Final Basin. June and November were selected as representative assessment months because they correspond to periods with relatively high and low constituent mass inventories within the Final Basin, thereby providing an indication of the potential range of event-scale impacts (Table 3.5). Across all assessed constituents, the predicted increase in mass remained less than 1% of the existing inventory. The event-based assessment therefore supports the conclusion that extreme rainfall events are unlikely to materially alter water quality within the receiving waterbody.

Table 3.4 Event-Based Pollutant Load Assessment

Constituent	MUSIC Concentration	Additional Volume (1% AEP)	Additional Mass (1% AEP)	Additional Volume (0.5% AEP)	Additional Mass (0.5% AEP)
TN	2.2 mg/L	30400 L	0.07 kg	37000L	0.08 kg
TP	200 µg/L	30400 L	0.01 kg	37000L	0.01 kg
TSS	77.6 mg/L	30400 L	2.36 kg	37000L	2.87 kg

Table 3.5 Mass increase percentage for June and November

Constituent	Assessment Month	Existing Monthly Median Mass (kg)	Additional Mass (1% AEP) (kg)	Increase (%)	Additional Mass (0.5% AEP) (kg)	Increase (%)
TN	June	67.73	0.07	0.1	0.08	0.12
TP	June	3.5	0.01	0.29	0.01	0.29
TSS	June	740.68	2.36	0.32	2.87	0.39
TN	November	39.95	0.07	0.18	0.08	0.2
TP	November	2.85	0.01	0.35	0.01	0.35
TSS	November	562.96	2.36	0.42	2.87	0.51

3.3.4 Overflow Assessment – Potential Transfer to Regatta

For the purpose of this screening-level assessment, water levels above 18.0 m AHD were assumed to represent water available for downstream transfer from the Final Basin to Regatta (Water Technology, 2026). Review of the monitoring record identified intermittent periods where water levels exceeded this threshold. The largest observed transfer event occurred during April-May 2024, when the estimated excess storage volume reached approximately 10.4 ML (Table 3.6). Analysis indicates that water transfers occur intermittently rather than continuously and are generally small relative to the overall storage volume of the Regatta system. The estimated transfer volumes and associated constituent loads were incorporated into the downstream Regatta assessment.

Table 3.6 Monthly median overflow volume to Regatta

Month	Monthly volume (ML)
Jan	1.13
Feb	1.35
April	10.4
May	3.10

3.3.5 Regatta Assessment

Figure 3.3 shows that the constituent mass transferred from the Final Basin during overflow events is small compared with the existing constituent mass inventory within Regatta. As a result, the overflow component is barely visible on the stacked bars, highlighting the limited influence of transferred loads on overall downstream water quality conditions.

The assessment indicates that water quality conditions within Regatta remain broadly consistent with existing conditions following inclusion of constituent loads transferred from the Final Basin.

Key findings are:

- TN concentrations continue to comply with the adopted WQO throughout the year.
- TSS concentrations remain below the adopted WQO throughout the year.
- TP compliance outcomes remain broadly unchanged.
- Existing seasonal Chl-a patterns remain unchanged.

Predicted increases in nutrient and suspended solid concentrations are negligible because the additional transferred loads are very small relative to the volume and existing constituent mass inventory of the Regatta system.

The results indicate that the proposed development is unlikely to materially alter water quality conditions or compliance outcomes within Regatta.

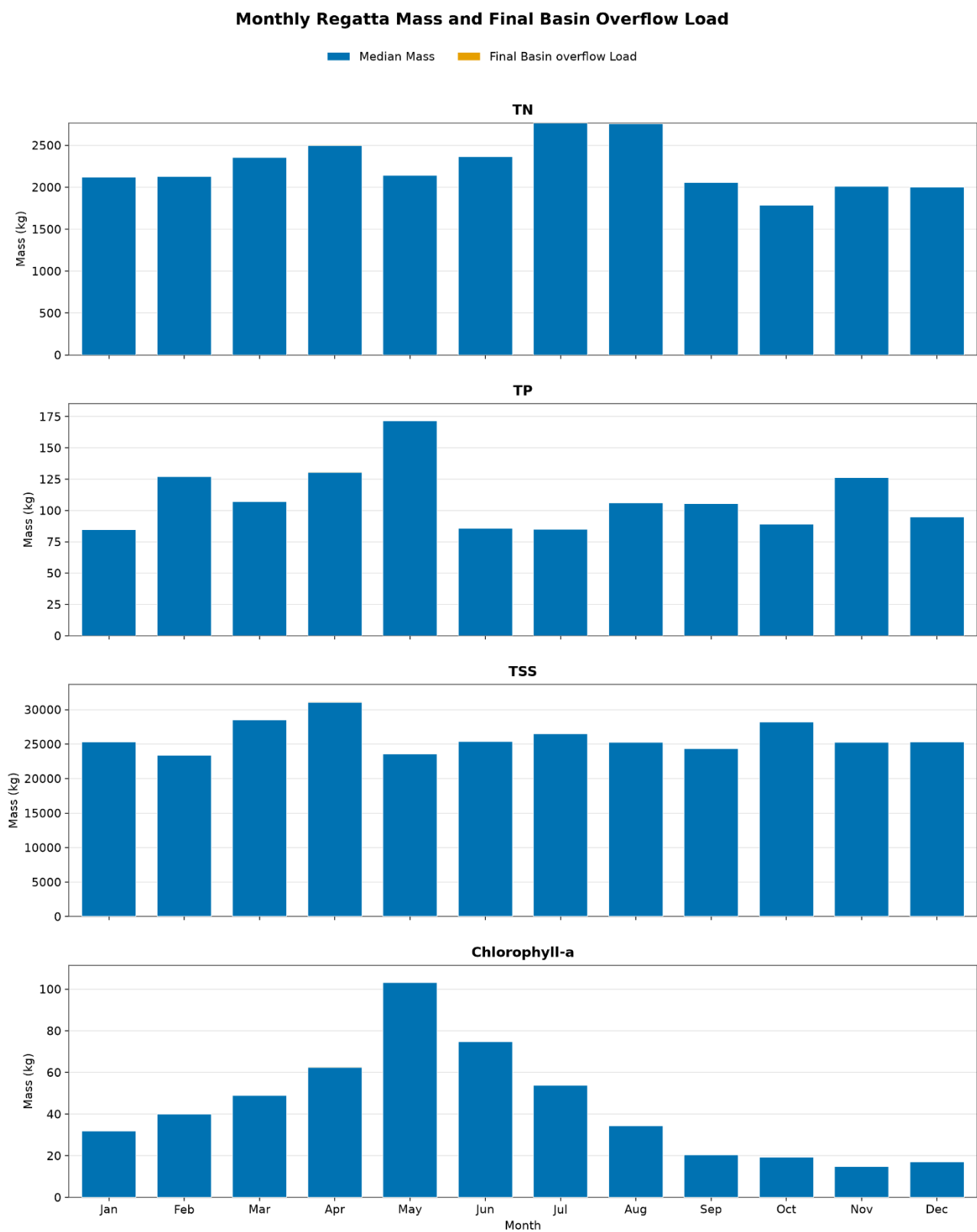


Figure 3.3 Post-development Regatta constituent mass inventory (Monthly median)

Table 3.7 Monthly median post-development concentrations in Regatta lake and comparison against adopted WQOs.

Month	TN (mg/L)	Criterion Met?	TP (µg/L)	Criterion Met?	TSS (mg/L)	Criterion Met?	Chl-a (µg/L)	Criterion Met?
Jan	0.50	Y	20.03	Y	5.98	Y	7.55	N
Feb	0.50	Y	30.00	N	5.52	Y	9.44	N
Mar	0.56	Y	25.32	N	6.74	Y	11.58	N
Apr	0.58	Y	30.48	N	7.25	Y	14.62	N
May	0.50	Y	39.99	N	9.99	Y	24.06	N
Jun	0.55	Y	20.04	Y	5.91	Y	17.45	N
Jul	0.65	Y	20.00	Y	6.24	Y	12.64	N
Aug	0.65	Y	25.03	N	5.97	Y	8.10	N
Sep	0.49	Y	25.02	N	5.77	Y	4.86	Y
Oct	0.43	Y	21.27	Y	6.71	Y	4.62	Y
Nov	0.48	Y	29.82	N	5.96	Y	3.50	Y
Dec	0.47	Y	22.50	Y	5.99	Y	4.04	Y

4 Discussion

4.1 Summary of Predicted Impacts

The assessment indicates that development-derived stormwater loads are small relative to the existing constituent mass inventories within both the Final Basin and Regatta. While minor increases in TN, TP and TSS loads are predicted under post-development conditions, the resulting changes in constituent concentrations are negligible and do not materially alter existing compliance outcomes.

The existing condition assessment indicates that water quality within the Final Basin is already characterised by elevated nutrient concentrations, with TP and Chl-a objectives exceeded during much or all of the year. The proposed development does not materially change these existing conditions, with post-development concentrations remaining broadly consistent with baseline conditions.

Event-based assessments for the 1% AEP and 0.5% AEP rainfall events indicate that development-derived loads represent less than 1% of the existing constituent mass inventory within the Final Basin. Similarly, constituent loads transferred to Regatta under assumed overflow conditions remain negligible relative to the existing volume and constituent mass inventory of the downstream lake system. Consequently, water quality conditions within Regatta are also predicted to remain broadly unchanged.

Key findings are summarised below in Table 4.1.

Table 4.1 Summary of Predicted Impacts

Assessment Component	Outcome
Final Basin TN	No material change
Final Basin TP	Existing exceedances remain unchanged
Final Basin TSS	Remains compliant
Final Basin Chl-a	No material change expected
Event-scale impacts	Less than 1% increase in constituent mass
Regatta TN	Remains compliant
Regatta TP	Compliance outcomes unchanged
Regatta TSS	Remains compliant
Regatta Chl-a	Existing seasonal patterns unchanged

4.2 Key Assumptions

The assessment is based on the following assumptions:

- Waterbody storage volumes can be estimated from available water level-storage relationships.
- Development-derived stormwater loads are represented by the available MUSIC modelling outputs.
- Water transferred from the Final Basin is assumed to be fully mixed with basin water prior to overflow.

- Water transferred from the Final Basin is assumed to have the same constituent concentration as the basin at the time of overflow.
- Water levels above 18.0 m AHD are assumed to represent excess storage available for transfer to the downstream lake system.
- Rowing Lake and Warm Up Lake (Regatta) are treated as a single downstream receptor.

These assumptions simplify the physical processes operating within the lake system and are considered appropriate for a screening-level mass-balance assessment.

4.3 Data Limitations

This assessment is based on available monitoring data and existing flood and MUSIC model outputs and should be considered a screening-level assessment. Key limitations include:

- Monitoring records contain temporal gaps, including missing water level data in both the Final Basin and Regatta datasets.
- Chlorophyll-a monitoring ceased in January 2025 and development-derived Chl-a loads were not available. Consequently, post-development Chl-a concentrations and algal responses could not be quantified directly.
- Stormwater runoff volumes and pollutant loads were derived from design modelling rather than measured stormwater data.
- MUSIC source loads were adopted, representing untreated runoff. The assessment therefore provides a conservative estimate of potential impacts and is likely to overestimate actual pollutant inputs to the Final Basin.
- Annual stormwater loads were distributed uniformly throughout the year, whereas actual runoff and pollutant loads will vary seasonally and between rainfall events.
- The simplified mass-balance approach assumes complete mixing within each waterbody and does not explicitly represent hydrodynamic processes, nutrient cycling, sediment interactions or spatial variability in water quality.
- Overflow volumes to Regatta were estimated using an adopted operational threshold of 18.0 m AHD and simplified assumptions regarding water transfers. Actual transfer volumes will depend on future hydrological conditions and lake operations.

Despite these limitations, the assessment provides a reasonable indication of the relative magnitude of development-derived loads compared with existing constituent masses within the Final Basin and Regatta.

5 Conclusion

The assessment indicates that the proposed development is unlikely to result in a material deterioration of water quality within the Final Basin or downstream Regatta. Existing conditions show that TP objectives are already exceeded in the Final Basin and periodically exceeded in Regatta Lake, while TN exceedances occur seasonally in the Final Basin and TSS generally complies with adopted objectives.

Development-derived TN, TP and TSS loads are small relative to the existing constituent mass inventories in both waterbodies, with event-based assessments indicating increases of less than 1% of existing constituent masses. The resulting increases in constituent concentrations are minor and do not materially alter existing seasonal water quality patterns.

Overflow assessments indicate that only small additional constituent loads are transferred from the Final Basin to the Regatta, resulting in negligible downstream water quality changes. Compliance outcomes against the adopted WQOs remain broadly consistent with existing conditions. Existing exceedances persist but are not materially worsened by the proposed development, while TSS is expected to continue to comply with adopted objectives.

Overall, the proposed development is not expected to materially increase nutrient enrichment risk or adversely affect the water quality of the Final Basin or Regatta. Given the conservative assumptions adopted, including the use of untreated MUSIC source loads, the assessment is considered to represent a precautionary estimate of potential impacts.

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